

HUEDIN, Romania

WMO: 150990

Lat: 46.8572N

Lon: 23.0325E

Elev: 561

StdP: 94.76

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.7	-11.4	-16.4	0.9	-11.8	-14.1	1.2	-10.5	9.6	3.3	7.3	2.2	0.4	210	0.484

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.6	29.5	19.7	27.8	19.3	26.2	18.7	21.0	27.2	20.2	26.2	19.4	25.1	2.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	14.7	23.8	18.1	14.0	23.3	17.3	13.3	22.4	63.6	27.1	60.7	26.5	57.9	25.2	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	6.1	5.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-2.8	0.7	4.1	9.3	14.5	18.0	18.9	19.3	14.7	9.5	3.3	-0.8
	DBStd	8.75	4.21	4.52	4.92	4.90	3.50	3.41	3.07	3.34	3.97	4.30	4.91	4.37
	HDD10.0	1533	396	261	191	69	7	0	0	0	6	62	207	336
	HDD18.3	3533	654	494	441	271	125	48	31	29	122	274	451	594
	CDD10.0	1206	0	1	8	49	146	239	277	288	146	47	6	1
	CDD18.3	164	0	0	0	1	7	37	49	59	12	0	0	0
	CDH23.3	1277	0	0	0	6	51	283	377	474	83	2	0	0
	CDH26.7	279	0	0	0	0	4	58	81	123	12	0	0	0
Wind	WSAvg	1.7	1.1	1.5	1.9	2.1	2.0	2.0	1.9	1.7	1.7	1.5	1.2	1.3
Precipitation	PrecAvg	712	35	33	41	53	77	106	97	63	67	50	46	48
	PrecMax	1004	60	78	110	97	147	203	196	117	166	122	107	92
	PrecMin	453	10	9	9	11	35	19	34	4	9	2	0	9
	PrecStd	134	14	16	24	20	31	44	42	27	34	30	22	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	14.1	19.9	24.0	26.8	30.3	30.9	31.7	28.8	23.2	17.7	13.2
		MCWB	6.6	8.1	11.5	14.1	16.8	21.0	20.0	20.2	18.0	14.4	12.0	9.4
	2%	DB	6.6	11.3	16.5	21.9	24.7	28.6	28.8	29.6	25.7	21.4	15.0	10.0
		MCWB	4.5	6.7	9.6	13.2	16.4	20.3	19.9	19.2	17.2	13.6	10.6	7.0
	5%	DB	4.2	9.3	13.5	19.5	22.9	26.6	26.9	27.6	23.8	19.2	12.6	7.5
		MCWB	2.8	5.9	8.5	12.1	15.9	19.4	19.3	19.2	16.5	12.9	9.5	5.6
	10%	DB	2.6	7.5	11.2	17.0	21.1	24.6	25.2	25.8	21.4	16.6	10.4	4.9
		MCWB	1.3	4.9	6.8	10.8	14.6	18.4	18.5	18.8	15.5	11.5	8.1	3.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	8.6	12.8	15.2	18.2	22.6	21.7	21.7	19.4	15.5	12.1	10.1
		MCDB	9.6	13.0	18.9	22.5	23.9	28.0	27.8	28.8	26.4	21.0	16.2	12.3
	2%	WB	5.0	7.3	10.5	14.0	17.4	20.9	20.8	20.8	18.0	14.3	11.1	7.5
		MCDB	6.8	10.3	15.5	20.7	23.1	26.9	26.4	27.4	24.1	20.3	14.8	9.9
	5%	WB	3.0	6.3	8.7	12.8	16.3	19.9	20.0	20.0	17.0	13.3	9.9	5.6
		MCDB	4.4	8.9	13.3	19.0	21.6	25.7	25.5	26.3	22.7	18.6	12.8	7.5
	10%	WB	1.7	5.1	7.3	11.4	15.3	18.9	19.2	19.2	16.0	12.0	8.5	3.9
		MCDB	2.5	7.6	11.1	16.2	20.2	24.1	24.3	25.1	20.9	16.3	10.8	5.0
Mean Daily Temperature Range	5% DB	MDBR	7.1	8.1	9.3	10.0	9.9	9.8	10.1	10.6	9.5	10.3	7.7	6.3
		MCDBR	8.2	10.5	12.7	13.0	12.0	12.2	11.9	12.7	12.7	14.1	10.1	9.0
		MCWBR	6.5	6.7	7.1	6.4	6.1	5.8	5.5	5.5	5.8	7.4	6.5	5.8
	5% WB	MCDBR	8.1	9.7	12.4	12.0	10.5	11.4	10.8	11.6	11.1	12.9	9.7	8.5
		MCWBR	5.9	6.5	7.2	6.2	6.0	5.9	5.5	5.4	5.4	7.1	6.2	5.8
Clear-Sky Solar Irradiance	taub	0.316	0.346	0.391	0.442	0.425	0.421	0.440	0.443	0.404	0.372	0.343	0.316	
	taud	2.306	2.244	2.173	2.073	2.184	2.234	2.189	2.174	2.249	2.317	2.342	2.338	
	Ebn at Noon	758	807	821	811	842	846	823	802	799	768	718	712	
	Edh at Noon	77	100	126	153	141	135	140	137	116	93	75	67	
All-Sky Solar Radiation	RadAvg	1.15	1.95	3.14	4.38	5.32	5.85	5.73	5.28	3.72	2.50	1.41	0.89	
	RadStd	0.16	0.27	0.46	0.46	0.45	0.52	0.45	0.51	0.49	0.43	0.30	0.20	0.12

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (59 neighbors)		+0.70	N/A	N/A	+0.99	N/A	N/A	-129	-195	+117	+58	

Nomenclature: See separate page